

SHRINITHI N

Aspiring Data Scientist

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🌐 [Shrinithi Nagarajan | LinkedIn](#)

PROFILE

Aspiring Data Scientist with a strong foundation in Python, machine learning, and data analysis. Experienced in building intelligent systems using libraries such as Pandas, NumPy, and Scikit-learn. Skilled in data preprocessing, exploratory data analysis, and predictive modeling. Passionate about applying data-driven solutions to solve real-world problems and continuously improving analytical and programming skills.

EDUCATION

B.E COMPUTER SCIENCE AND ENGINEERING <i>Sri Shakthi Institute of Engineering and Technology</i> CGPA – 7.89	2023 – Present
HIGHER SECONDARY CERTIFICATE <i>Thiyagi NG Ramaswamy Memorial and Higher Secondary School</i> HSC – 83%	2022 - 2023
SECONDARY SCHOOL LEAVING CERTIFICATE <i>Thiyagi NG Ramaswamy Memorial and Higher Secondary School</i> SSLC – 100%	2020 - 2021

PROJECTS

Powerloop – Smart Campus Energy Optimization System

Developed an intelligent energy management system to optimize electricity usage in campus environments.

- Implemented real-time human detection using YOLOv8 and OpenCV to identify room occupancy.
- Integrated ESP32 microcontroller to automatically control electrical relays based on occupancy status.
- Built a web dashboard using Flask, HTML, CSS, and JavaScript to monitor room activity and energy usage.
- Enabled automated power optimization to reduce unnecessary energy consumption in classrooms.

AI Similar Feature Detection on PCB Boards – Python

Developed an automated PCB component similarity detection system using DINOv2 and Computer Vision techniques.

- Implemented sliding window algorithms and cosine similarity calculations to accurately identify and match similar PCB components.
- Designed bounding box visualization for detected components, gaining practical exposure to Deep Learning and industrial inspection systems.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Analysis & Tools: Pandas, NumPy, Matplotlib, Seaborn

Machine Learning / AI: Scikit-learn, TensorFlow

Tools and Platform: VS Code, PyCharm, Git, GitHub

CERTIFICATIONS

- Python for Data Science – Udemy
- Python for Data Science Bootcamp - Udemy

ACHIEVEMENTS

- Secured 3rd Place in the National Level Project Expo “Tech Vision 2026” at CMR Institute of Technology, Bangalore for the project “Powerloop – Smart Campus Energy Optimization System”.
- Participated in the Smart India Hackathon (SIH) for two consecutive years, working in teams to develop innovative technology solutions for real-world challenges.